

Movie Streaming Platform Database **(StreamHub)**

Team Details

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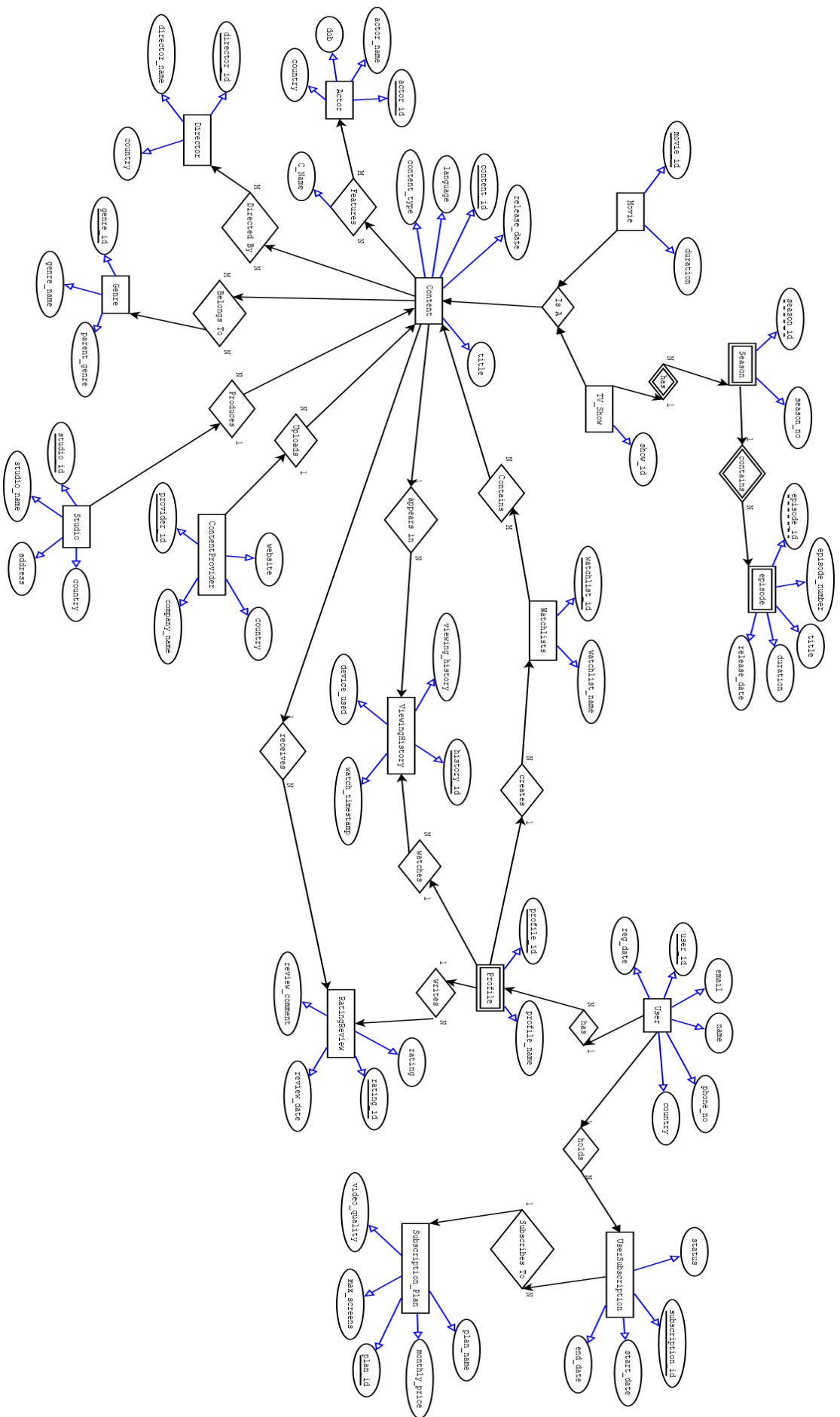
Group Representative

Name: Kaushal Dodia

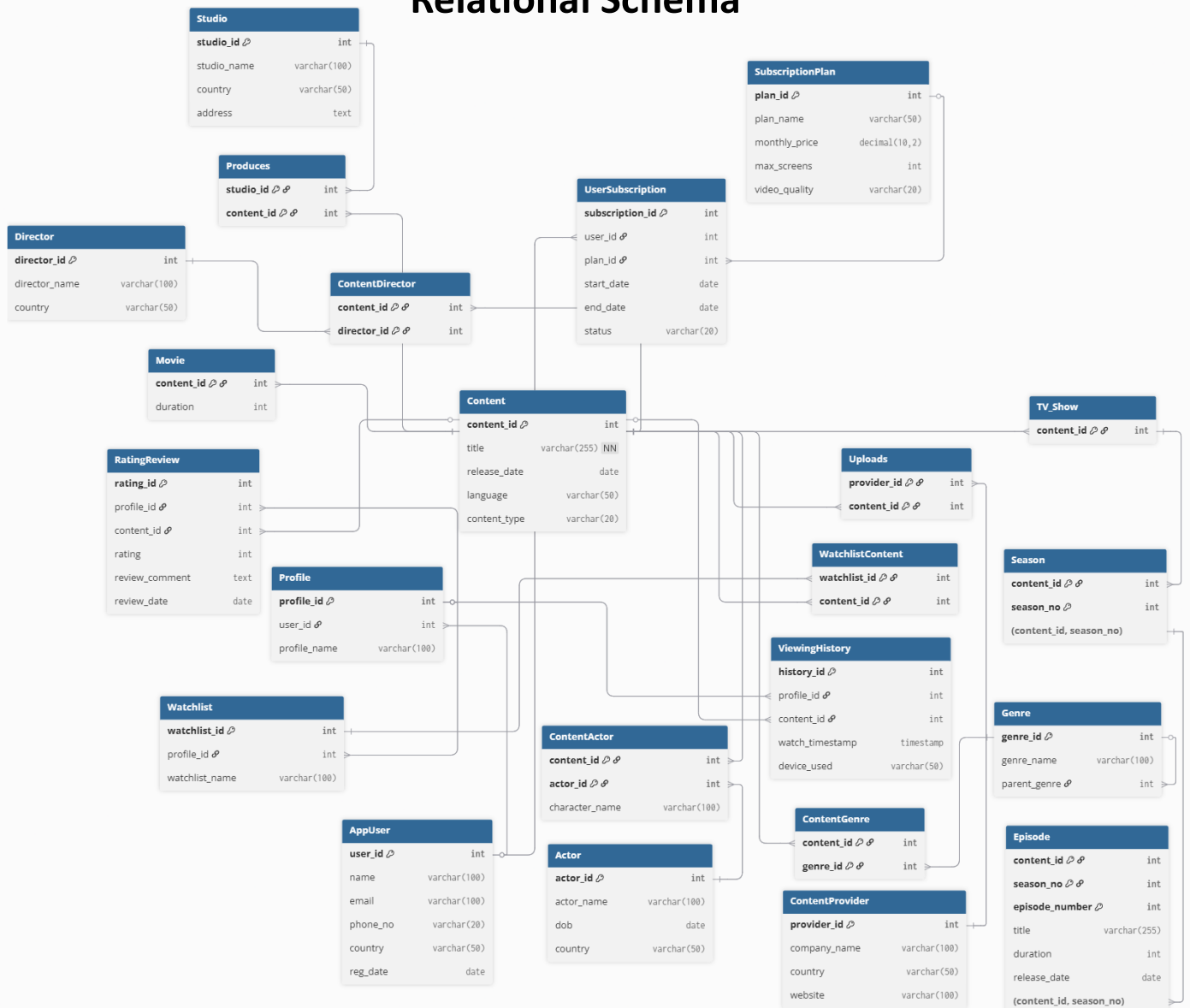
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ERD



Relational Schema



Credit: <https://dbdiagram.io>

BCNF Criteria Definition

A relation is in **BCNF (Boyce-Codd Normal Form)** if and only if, for every non-trivial functional dependency $X \rightarrow Y$, X is a **superkey (or candidate key)** of the relation.

1. Core Content Entities

Content

- i. **Candidate Key(s):** content_id
- ii. **Functional Dependencies:**
 $content_id \rightarrow title, release_date, language, content_type$
- iii. **BCNF Status:** Yes
 - The only determinant is content_id, which is the primary key.

Movie

- i. **Candidate Key(s):** content_id
- ii. **Functional Dependencies:**
 $content_id \rightarrow duration$
- iii. **BCNF Status:** Yes
 - Determinant is the primary key.

TV_Show

- i. **Candidate Key(s):** content_id
- ii. **Functional Dependencies:**
None (only trivial dependency: $content_id \rightarrow content_id$)
- iii. **BCNF Status:** Yes

Season

- i. **Candidate Key(s):** (content_id, season_no)
- ii. **Functional Dependencies:**
None (non-trivial)
- iii. **BCNF Status:** Yes

Episode

- i. **Candidate Key(s):** (content_id, season_no, episode_number)
- ii. **Functional Dependencies:**
 $(content_id, season_no, episode_number) \rightarrow title, duration, release_date$
- iii. **BCNF Status:** Yes
 - Determinant is the primary key.
 - No partial dependencies exist because attributes depend on the full composite key.

2. User & Subscription Entities

AppUser

- i. **Candidate Key(s):** user_id, email
- ii. **Functional Dependencies:**
 $user_id \rightarrow name, email, phone_no, country, reg_date$
 $email \rightarrow user_id, name, phone_no, country, reg_date$
- iii. **BCNF Status:** Yes
 - email has a UNIQUE constraint, making it a candidate key alongside user_id.

Profile

- i. **Candidate Key(s):** profile_id
- ii. **Functional Dependencies:**
 $profile_id \rightarrow user_id, profile_name$
- iii. **BCNF Status:** Yes

SubscriptionPlan

- i. **Candidate Key(s):** plan_id
- ii. **Functional Dependencies:**
 $plan_id \rightarrow plan_name, monthly_price, max_screens, video_quality$
- iii. **BCNF Status:** Yes

UserSubscription

- i. **Candidate Key(s):** subscription_id
- ii. **Functional Dependencies:**
 $subscription_id \rightarrow user_id, plan_id, start_date, end_date, status$
- iii. **BCNF Status:** Yes

3. Interaction Entities

RatingReview

- i. **Candidate Key(s):** $rating_id$
- ii. **Functional Dependencies:**
 $rating_id \rightarrow profile_id, content_id, rating, review_comment, review_date$
- iii. **BCNF Status:** Yes

ViewingHistory

- i. **Candidate Key(s):** $history_id$
- ii. **Functional Dependencies:**
 $history_id \rightarrow profile_id, content_id, watch_timestamp, device_used$
- iii. **BCNF Status:** Yes

Watchlist

- i. **Candidate Key(s):** $watchlist_id$
- ii. **Functional Dependencies:**
 $watchlist_id \rightarrow profile_id, watchlist_name$
- iii. **BCNF Status:** Yes

4. Metadata & Production Entities

Actor

Functional Dependencies:
 $actor_id \rightarrow actor_name, dob, country$

Director

Functional Dependencies:
 $director_id \rightarrow director_name, country$

Genre

Functional Dependencies:
 $genre_id \rightarrow genre_name, parent_genre$

Studio

Functional Dependencies:
 $studio_id \rightarrow studio_name, country, address$

ContentProvider

Functional Dependencies:

$provider_id \rightarrow company_name, country, website$

BCNF Status (All Above): Yes

- Each uses a surrogate key that determines all attributes.

5. Many-to-Many (M:N) Mapping Entities

ContentActor

- Candidate Key(s):** (content_id, actor_id)
- Functional Dependencies:**
 $(content_id, actor_id) \rightarrow character_name$
- BCNF Status:** Yes
 - character_name depends on the full composite key.

ContentDirector, ContentGenre, WatchlistContent, Produces, Uploads

- Functional Dependencies:**
None (non-trivial)
- BCNF Status:** Yes
 - These tables only contain composite keys for relationships.

All relations satisfy **BCNF**, as every determinant is a candidate key or superkey, and no partial or transitive dependency violates the BCNF condition.